



BOARD QUESTION PAPER: FEBRUARY 2025

CHEMISTRY

Time: 3 Hrs.

Max. Marks: 70

General Instructions:

The question paper is divided into **four** sections.

- (1) **Section A:** Q. No. 1 contains **Ten** multiple choice type of questions carrying **One** mark each. Only the first attempt will be considered for evaluation.
Q. No. 2 contains **Eight** very short answer type of questions carrying **One** mark each.
- (2) **Section B:** Q. No. 3 to Q. No. 14 are **Twelve** short answer type-I questions carrying **Two** marks each. (Attempt **any Eight**)
- (3) **Section C:** Q. No. 15 to Q. No. 26 are **Twelve** short answer type-II questions carrying **Three** marks each. (Attempt **any Eight**)
- (4) **Section D:** Q. No. 27 to Q. No. 31 are **Five** long answer type of questions carrying **Four** marks each. (Attempt **any Three**)
- (5) Use of log table is allowed. Use of calculator is not allowed.
- (6) Figures to the right indicate full marks.
- (7) Given data:
 - i. $R = 8.314 \text{ J/K/mol}$
 - ii. Atomic mass of Na = 23
 - iii. K_f for water = $1.86 \text{ K kg mol}^{-1}$
 - iv. $1F = 96500C$
 - v. $N_A = 6.022 \times 10^{23}$

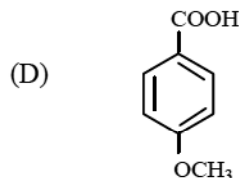
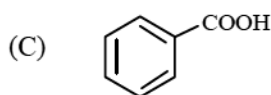
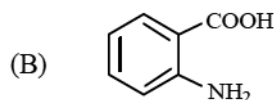
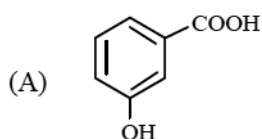
SECTION – A

Q.1. Select and write the correct answers for the following multiple choice type of questions: [10]

- i. Schottky defect is NOT observed in _____.
(A) NaCl (B) KCl (C) AgBr (D) NiO
- ii. The freezing point of 0.1m aqueous solution of urea, if K_f for water is $1.86 \text{ K kg mol}^{-1}$ is _____.
(A) 1.86°C (B) -1.86°C (C) 0.186°C (D) -0.186°C
- iii. Ozone layer is depleted by _____.
(A) NO (B) NO_2 (C) NO_3 (D) N_2O_5
- iv. When excess of AgNO_3 is added to a complex, one mole of AgCl is precipitated. The formula of complex is _____.
(A) $[\text{CoCl}_2(\text{NH}_3)_4]\text{Cl}$ (B) $[\text{CoCl}(\text{NH}_3)_5]\text{Cl}_2$
(C) $[\text{CoCl}_3(\text{NH}_3)_3]$ (D) $[\text{Co}(\text{NH}_3)_6]\text{Cl}_3$
- v. The value of Δn_g for the oxidation of 4 mole of sulphur dioxide to sulphur trioxide is _____.
(A) -2 (B) 2 (C) -4 (D) 4
- vi. One dimensional nanostructure amongst the following is _____.
(A) Nanoparticles (B) Nanotubes
(C) Nanofilms (D) Nanorods
- vii. Which formula co-relates degree of dissociation and concentration of electrolyte?
(A) $c = \sqrt{\frac{K_a}{\alpha}}$ (B) $\alpha = \sqrt{\frac{K_a}{c}}$
(C) $c = \sqrt{K_a \alpha}$ (D) $c = \sqrt{\frac{\alpha}{K_a}}$



viii. The highest acidic compound among the following is _____.



ix. The formula used to calculate molar conductivity of an electrolyte is _____.

(A) $\Lambda = \frac{1000c}{k}$

(B) $c = \frac{1000\Lambda}{k}$

(C) $\Lambda = \frac{1000k}{c}$

(D) $k = \frac{1000}{\Lambda c}$

x. Which of the following is a secondary amine?

(A) Cyclohexylamine

(B) Isopropylamine

(C) Diphenylamine

(D) N, N-Dimethylaniline

Q.2. Answer the following questions:

[8]

- Write the structural formula of N, N-dimethylethanamine.
- Write the reagents used for the reduction of carbonyl group in Clemmensen's reduction.
- Write the IUPAC name of isoprene.
- The rate law equation for $A \rightarrow \text{Product}$, is $\text{rate} = k[A]^x$. What is the effect of increase in concentration of 'A' on rate of reaction, if $x < 0$?
- What is the molality of an aqueous solution of KBr having freezing point -3.72°C (K_f for water is $1.86 \text{ K kg mol}^{-1}$)?
- Write the balanced chemical equation, when excess of ammonia is treated with chlorine
- Write the number of donor atoms present in EDTA, during formation of complex.
- Write the names of the metal elements in brass alloy.

SECTION – B

Attempt any EIGHT of the following questions:

[16]

Q.3. Derive the relation between half-life and rate constant for a first order reaction.

Q.4. a. State Henry's law.

b. Define: Osmotic pressure

Q.5. Write the differences between lanthanoids and actinoids.

Q.6. Write anomalous behaviour of oxygen with respect to:

i. Atomicity

ii. Oxidation state

iii. Magnetic property

iv. Nature of hydrides



- Q.7. What is the action of:
- Liquid bromine in acetic acid on anisole
 - Soda-lime on sodium acetate?
- Q.8. Calculate the work done in kJ in a reaction, if volume of the reactant decreases from 8 dm^3 to 4 dm^3 against 43 bar pressure. [$1 \text{ dm}^3 \cdot \text{bar} = 100 \text{ J}$]
- Q.9. Explain ionization isomers with suitable example in complexes.
- Q.10. Write preparation of glucose from sucrose.
- Q.11. How many coulombs of electricity is required to produce 1g of sodium metal by reduction of sodium ion?
- Q.12. Write the structural formula and IUPAC name of the alcohol having molecular formula $\text{C}_4\text{H}_{10}\text{O}$ which does not undergo oxidation under normal condition.
- Q.13. Identify 'A' and 'B' in the following reaction and rewrite the complete reaction:
- $$\text{CH}_3 - \text{CH} = \text{CH}_2 \xrightarrow[\text{Peroxide}]{\text{HBr}} \text{A} \xrightarrow[\text{-KBr}]{\text{alcoholic KCN}} \text{B}$$
- Q.14. Write the reaction for the preparation of:
- acetaldehyde by Rosenmund reaction.
 - benzaldehyde by Gatterman-Koch formylation.

SECTION – C

Attempt any EIGHT of the following questions:

[24]

- Q.15. Write the general electronic configuration of 3d series. Draw the structures of sulphuric acid and thiosulphuric acid.
- Q.16. Define conjugate acid-base pair. The hydroxyl ion concentration in aqueous solution of NaOH is $2 \times 10^{-4} \text{ mol dm}^{-3}$. Calculate pH of the solution.
- Q.17. What is atom economy? Explain any two applications of nanomaterials.
- Q.18. What is peptide bond? How is it formed? Write the name and formula of the reagent used to convert alkylhalide to nitroalkane.
- Q.19. a. Write the reactions for the action of following reagents on phenol:
- Nitrating mixture
 - Zinc dust
- b. What is the action of phosphorous pentachloride on ethyl methyl ether?
- Q.20. a. Write the formula to calculate EAN.
- b. Explain formation of $[\text{CO}(\text{NH}_3)_6]^{3+}$ complex ion with respect to:
- Type of hybridisation
 - Magnetic property
- Q.21. a. Calculate spin only magnetic moment of M^{2+} ion.
[atomic number of M = 26]
- b. Write condensed electronic configuration of Gadolinium.
[Z = 64]
- Q.22. a. Write the reducing agents used to convert Fe_2O_3 to 'Fe' in the reduction zone of blast furnace.
- b. Write chemical equations involved in:
- Carbylamine reaction for ethylamine.
 - Hoffmann Bromamide degradation for acetamide



- Q.23. a. Explain Cannizzaro's reaction with the help of benzaldehyde.
b. Write the reaction for the conversion of cyclohexene to adipic acid.

- Q.24. a. Define zero order reaction.

A reaction takes place in two steps:

- i. $\text{NO}_{(g)} + \text{Cl}_{2(g)} \longrightarrow \text{NOCl}_{2(g)}$
ii. $\text{NOCl}_{2(g)} + \text{NO}_{(g)} \longrightarrow 2\text{NOCl}_{(g)}$

Write the overall reaction and identify the reaction intermediate.

- Q.25. ΔH for formation of ethane gas is -84.4 kJ at 300 K . Calculate ΔU for the reaction.

- Q.26. Mention the types of polymers formed on the basis of intermolecular forces. Write any two uses of low density polyethylene.

SECTION – D

Attempt any THREE of the following questions:

[12]

- Q.27. a. An element with molar mass 27 g/mol forms a cubic unit cell with edge length 405 pm . If density of the crystal is 2.7 g cm^{-3} , identify the type of unit cell.
b. Derive the equation of Raoult's law for binary solution containing non-volatile solute.
- Q.28. a. State whether entropy change is positive or negative in the following examples:
i. Melting of ice
ii. Vaporisation of a liquid
b. Explain 'common ion effect' with example.
- Q.29. Draw a neat and labelled diagram of a lead accumulator cell. Write the overall reactions taking place at cathode and anode during discharging of the cell.
- Q.30. a. Define a unit cell.
Which colour is shown by NaCl crystal due to formation of F-centre?
b. Why does fluorine show anomalous behaviour in '17 group' elements?
- Q.31. a. Write salient features of SN^2 mechanism.
b. What is the action of following reagents on bromomethane:
i. bromobenzene
ii. mercurous fluoride